

A PROJECT TITLE

ON

Traffic Flow Data Visualization Using PCA and t-SNE

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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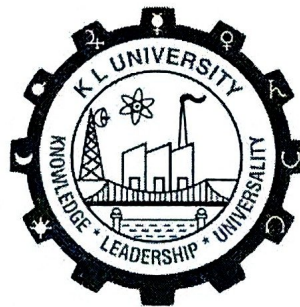
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DEPARTMENT OF CSE

Name of Title: Traffic Flow Data Visualization Using PCA and t-SNE

1. Abstract:

Traffic congestion has become a significant challenge in modern cities due to rapid urbanization, population growth, and the increasing number of vehicles on the road. Heavy traffic not only increases travel time but also leads to higher fuel consumption, air pollution, and economic losses. Effective analysis of traffic flow data is essential for understanding congestion patterns and improving transportation systems. However, traffic datasets are often high-dimensional, containing multiple features such as vehicle speed, traffic volume, occupancy, travel time, and road conditions. Visualizing and interpreting such complex data can be difficult using traditional methods.

This project presents Traffic Flow Data Visualization Using Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbor Embedding (t-SNE) to analyze and visualize traffic flow data efficiently. PCA is a linear dimensionality reduction technique that transforms high-dimensional data into a smaller set of principal components while preserving most of the data's variance. It helps identify the most influential traffic features and provides a simplified representation of the dataset. On the other hand, t-SNE is a non-linear dimensionality reduction technique that focuses on preserving local relationships between data points, making it highly effective for identifying clusters and hidden patterns in traffic conditions.

The project compares the visualizations generated by PCA and t-SNE to understand different traffic states, including free-flowing, moderate, and congested traffic. The resulting visual representations enable transportation planners, researchers, and traffic authorities to identify congestion hotspots, detect abnormal traffic patterns, and support data-driven decision-making for traffic management. While PCA provides a computationally efficient overview of the dataset, t-SNE offers more detailed and intuitive visualization of complex traffic behavior.

Overall, this project demonstrates how dimensionality reduction techniques can transform complex traffic datasets into meaningful visual insights. The proposed approach supports better traffic analysis, enhances understanding of congestion patterns, and contributes to the development of smarter and more efficient intelligent transportation systems for future urban mobility.

2. INTRODUCTION:

Traffic congestion has become one of the most critical challenges in urban transportation systems due to rapid population growth, urbanization, and the increasing number of vehicles on roads. Congested roads result in longer travel times, increased fuel consumption, higher carbon emissions, and economic losses. Efficient traffic management requires analyzing large volumes of traffic flow data collected from sensors, cameras, GPS devices, and intelligent transportation systems. These datasets contain multiple attributes such as vehicle speed, traffic density, traffic volume, occupancy, and travel time, making them complex and difficult to interpret using conventional visualization techniques.

Data visualization plays a vital role in understanding traffic patterns and identifying congestion-prone areas. However, high-dimensional traffic data cannot be effectively represented in its original form. Dimensionality reduction techniques help simplify complex datasets while preserving important information, allowing meaningful visualization and analysis. Among these techniques, Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbor Embedding (t-SNE) are widely used for exploring high-dimensional data.

PCA is a linear dimensionality reduction method that transforms the original dataset into a smaller number of principal components while retaining most of the data's variance. It helps identify the key factors influencing traffic flow and provides a computationally efficient overview of the dataset. In contrast, t-SNE is a non-linear dimensionality reduction technique designed to preserve local relationships between data points. It produces clear visual clusters, making it easier to distinguish different traffic conditions such as free-flowing, moderate, and heavily congested traffic.

This project focuses on visualizing traffic flow data using PCA and t-SNE to gain better insights into traffic behavior. By comparing the performance of these two techniques, the project aims to identify congestion patterns, detect hidden structures within the data, and support effective traffic management. The visualization results can assist researchers, transportation planners, and government authorities in making informed decisions for reducing congestion and improving road network efficiency. This study also demonstrates the importance of dimensionality reduction techniques in developing intelligent transportation systems and promoting smarter urban mobility.

3. SOFTWARE REQUIREMENTS:

The following software components are required to develop and implement the Traffic Flow Data Visualization Using PCA and t-SNE project.

1. Operating System

- Windows 10/11, Ubuntu 20.04 or later, or macOS

2. Programming Language

- Python 3.10 or above

3. Development Environment (IDE)

- Visual Studio Code (VS Code)
- Jupyter Notebook / JupyterLab
- Google Colab (Optional)

4. Python Libraries

- NumPy – Numerical computations and array operations
- Pandas – Data loading, preprocessing, and analysis
- Matplotlib – Data visualization and plotting
- Seaborn – Statistical data visualization
- Scikit-learn – Implementation of PCA, t-SNE, preprocessing, and evaluation

5. Dataset

- Traffic Flow Dataset (CSV format)
- Features may include:
 - Vehicle Speed
 - Traffic Volume
 - Traffic Density
 - Occupancy
 - Travel Time
 - Road Condition (if available)

6. Data Storage

- CSV files
- Local file system

7. Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: At least 2 GB free disk space
- Internet: Required only for downloading datasets and installing libraries

8. Software Dependencies

Install the required Python packages using:

```
pip install numpy pandas matplotlib seaborn scikit-learn
```

9. Output

- PCA-based 2D visualization of traffic flow data
- t-SNE-based 2D visualization of traffic flow data
- Comparative analysis of PCA and t-SNE visualizations
- Identification of traffic congestion patterns and clusters



Signature of the Guide



Course Instructor